

2004 年数学(三)真题解析

一、填空题

(1)【答案】 $1, -4$.

【解】 由 $\lim_{x \rightarrow 0} \frac{\sin x}{e^x - a}(\cos x - b) = 5$, 得 $\lim_{x \rightarrow 0} (e^x - a) = 0$, 于是 $a = 1$;

再由 $5 = \lim_{x \rightarrow 0} \frac{\sin x}{e^x - 1}(\cos x - b) = \lim_{x \rightarrow 0} (\cos x - b) = 1 - b$, 得 $b = -4$.

(2)【答案】 $-\frac{g'(v)}{g^2(v)}$.

【解】 令 $u = xg(y)$, $v = y$, 则 $x = \frac{u}{g(v)}$, $y = v$,

于是 $f(u, v) = \frac{u}{g(v)} + g(v)$.

由 $\frac{\partial f}{\partial u} = \frac{1}{g(v)}$, 得 $\frac{\partial^2 f}{\partial u \partial v} = -\frac{g'(v)}{g^2(v)}$.

(3)【答案】 $-\frac{1}{2}$.

【解】
$$\begin{aligned} \int_{\frac{1}{2}}^2 f(x-1) dx &= \int_{\frac{1}{2}}^2 f(x-1) d(x-1) = \int_{-\frac{1}{2}}^1 f(x) dx \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} x e^{x^2} dx + \int_{\frac{1}{2}}^1 (-1) dx = \int_{\frac{1}{2}}^1 (-1) dx = -\frac{1}{2}. \end{aligned}$$

(4)【答案】 2 .

【解】
$$\begin{aligned} f(x_1, x_2, x_3) &= (x_1 + x_2)^2 + (x_2 - x_3)^2 + (x_3 + x_1)^2 \\ &= 2x_1^2 + 2x_2^2 + 2x_3^2 + 2x_1x_2 + 2x_1x_3 - 2x_2x_3, \end{aligned}$$

则二次型 $f(x_1, x_2, x_3)$ 的矩阵为 $A = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}$,

由 $A \rightarrow \begin{pmatrix} 1 & 2 & -1 \\ 0 & -3 & 3 \\ 0 & 0 & 0 \end{pmatrix}$, 得 $r(A) = 2$, 即二次型 $f(x_1, x_2, x_3)$ 的秩为 2 .

(5)【答案】 $\frac{1}{e}$.

【解】 由 $X \sim E(\lambda)$, 得 $D(X) = \frac{1}{\lambda^2}$, 且其分布函数为

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0, \end{cases}$$

于是 $P\{X > \sqrt{D(X)}\} = P\left\{X > \frac{1}{\lambda}\right\} = 1 - P\left\{X \leq \frac{1}{\lambda}\right\} = 1 - F\left(\frac{1}{\lambda}\right) = \frac{1}{e}$.

方法点评: 本题考查指数分布的数字特征及分布函数的概念.

重要的随机变量及其数字特征有:

(1)二项分布

设 $X \sim B(n, p)$, 则 $E(X) = np$, $D(X) = npq$.

(2)泊松分布

设 $X \sim P(\lambda)$, 则 $E(X) = \lambda$, $D(X) = \lambda$.

(3)指数分布

设 $X \sim E(\lambda)$, 则 $E(X) = \frac{1}{\lambda}$, $D(X) = \frac{1}{\lambda^2}$,

且密度函数及分布函数为

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad F(x) = \begin{cases} 1 - e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

(4)正态分布

设 $X \sim N(\mu, \sigma^2)$, 则 $E(X) = \mu$, $D(X) = \sigma^2$.

(6)【答案】 σ^2 .

$$\text{【解】} \quad \frac{\sum_{i=1}^{n_1} (X_i - \bar{X})^2}{\sigma^2} \sim \chi^2(n_1 - 1), \quad \frac{\sum_{j=1}^{n_2} (Y_j - \bar{Y})^2}{\sigma^2} \sim \chi^2(n_2 - 1),$$

$$\text{由 } E \left[\frac{\sum_{i=1}^{n_1} (X_i - \bar{X})^2}{\sigma^2} \right] = n_1 - 1, \quad E \left[\frac{\sum_{j=1}^{n_2} (Y_j - \bar{Y})^2}{\sigma^2} \right] = n_2 - 1, \text{ 得}$$

$$E \left[\sum_{i=1}^{n_1} (X_i - \bar{X})^2 \right] = (n_1 - 1)\sigma^2, \quad E \left[\sum_{j=1}^{n_2} (Y_j - \bar{Y})^2 \right] = (n_2 - 1)\sigma^2,$$

$$\text{于是 } E \left[\frac{\sum_{i=1}^{n_1} (X_i - \bar{X})^2 + \sum_{j=1}^{n_2} (Y_j - \bar{Y})^2}{n_1 + n_2 - 2} \right] = \frac{(n_1 - 1)\sigma^2 + (n_2 - 1)\sigma^2}{n_1 + n_2 - 2} = \sigma^2.$$

二、选择题

(7)【答案】 (A).

$$\text{【解】} \quad |f(x)| = \left| \frac{\sin(x-2)}{(x-1)(x-2)^2} \right|,$$

$$\text{当 } x \in (-1, 0) \text{ 时, } |f(x)| = \left| \frac{\sin(x-2)}{(x-1)(x-2)^2} \right| \leq \frac{1}{|(x-1)(x-2)^2|} < \frac{1}{2}, \text{ 应选(A).}$$

(8)【答案】 (D).

$$\text{【解】} \quad \lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} f\left(\frac{1}{x}\right) = \lim_{t \rightarrow \infty} f(t) = a, \quad g(0) = 0,$$

当 $a = 0$ 时, $g(x)$ 在 $x = 0$ 处连续;

当 $a \neq 0$ 时, $x = 0$ 为 $g(x)$ 的可去间断点,

即 $g(x)$ 在 $x = 0$ 处连续与否与 a 的取值有关, 应选(D).

(9)【答案】 (C).

$$\text{【解】 方法一} \quad f(x) = \begin{cases} x^2 - x, & x \leq 0, \\ x - x^2, & 0 < x < 1, \\ x^2 - x, & x \geq 1, \end{cases} \quad f'(x) = \begin{cases} 2x - 1, & x < 0, \\ 1 - 2x, & 0 < x < 1, \\ 2x - 1, & x > 1, \end{cases}$$

$$f''(x) = \begin{cases} 2, & x < 0, \\ -2, & 0 < x < 1, \\ 2, & x > 1, \end{cases}$$

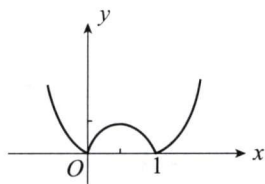
当 $x < 0$ 时, $f'(x) = 2x - 1 < 0$;

当 $0 < x < \frac{1}{2}$ 时, $f'(x) = 1 - 2x > 0$, 则 $x = 0$ 为 $f(x)$ 的极小值点;

又当 $x < 0$ 时, $f''(x) = 2 > 0$; 当 $0 < x < 1$ 时, $f''(x) = -2 < 0$,

则 $(0, 0)$ 为曲线 $y = f(x)$ 的拐点, 应选(C).

方法二 函数 $f(x) = |x(1-x)|$ 的图像如右图所示, 显然 $x=0$ 为 $f(x)$ 的极值点, $(0, 0)$ 为曲线 $y = f(x)$ 的拐点, 应选(C).



二(9)题图

(10)【答案】 (B).

【解】 取 $u_n = (-1)^n$, 显然 $\sum_{n=1}^{\infty} (u_{2n-1} + u_{2n})$ 收敛, 但 $\sum_{n=1}^{\infty} u_n$ 发散, ①不对;

因为增加、减少、改变级数的有限项不改变级数的敛散性, 所以②正确;

由 $\lim_{n \rightarrow \infty} \frac{u_{n+1}}{u_n} > 1$, 得 $\lim_{n \rightarrow \infty} u_n \neq 0$, 由级数收敛的必要条件得 $\sum_{n=1}^{\infty} u_n$ 发散, ③正确;

取 $u_n = \frac{1}{n^2} + \frac{1}{n}$, $v_n = \frac{1}{n^2} - \frac{1}{n}$, 显然 $\sum_{n=1}^{\infty} (u_n + v_n)$ 收敛, 但 $\sum_{n=1}^{\infty} u_n$ 与 $\sum_{n=1}^{\infty} v_n$ 都发散, ④不对, 应选(B).

(11)【答案】 (D).

【解】 $f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} > 0$, 由极限保号性, 存在 $\delta_1 > 0$, 当 $x \in (a, a + \delta_1)$ 时,

$\frac{f(x) - f(a)}{x - a} > 0$, 从而 $f(x) > f(a)$, 于是存在 $x_0 \in (a, b)$, 使得 $f(x_0) > f(a)$;

$f'(b) = \lim_{x \rightarrow b} \frac{f(x) - f(b)}{x - b} < 0$, 由极限保号性, 存在 $\delta_2 > 0$, 当 $x \in (b - \delta_2, b)$ 时,

$\frac{f(x) - f(b)}{x - b} < 0$, 从而 $f(x) > f(b)$, 于是存在 $x_0 \in (a, b)$, 使得 $f(x_0) > f(b)$,

则(A), (B)正确;

因为 $f'(x)$ 连续且 $f'(a)f'(b) < 0$, 所以由零点定理, 存在 $x_0 \in (a, b)$, 使得 $f'(x_0) = 0$, 则(C)正确, 应选(D).

方法点评: 本题考查极限保号性与零点定理.

关于函数在一点导数的符号有如下结论:

(1) 若 $f'(a) > 0$, 则存在 $\delta > 0$, 使得 $\begin{cases} f(x) < f(a), & x \in (a - \delta, a), \\ f(x) > f(a), & x \in (a, a + \delta); \end{cases}$

(2) 若 $f'(a) < 0$, 则存在 $\delta > 0$, 使得 $\begin{cases} f(x) > f(a), & x \in (a - \delta, a), \\ f(x) < f(a), & x \in (a, a + \delta); \end{cases}$

(3) 若 $f'(a) > 0$, 则函数 $f(x)$ 在 $x = a$ 的邻域内不一定单调增加, 如:

$$f(x) = \begin{cases} \frac{x}{2} + x^2 \sin \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases} \quad f'(0) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \frac{1}{2} > 0,$$

$$f'(x) = \begin{cases} \frac{1}{2} - \cos \frac{1}{x}, & x \neq 0, \\ 0, & x = 0, \end{cases} \quad \text{显然 } f(x) \text{ 在 } x=0 \text{ 的邻域内不单调.}$$

(12)【答案】 (D).

【解】 方法一 因为矩阵 A, B 等价, 所以根据矩阵等价的定义, 存在可逆矩阵 P, Q , 使得 $PAQ=B$, 于是当 $|A|=0$ 时, $|B|=0$, 应选(D).

方法二 矩阵 A, B 等价的充分必要条件为 $r(A)=r(B)$, 所以当 $|A|=0$ 时, $r(A)<n$, 于是 $r(B)<n$, 故 $|B|=0$, 应选(D).

(13)【答案】 (B).

【解】 由非齐次线性方程组 $AX=b$ 有解但不唯一, 得 $r(A)<n$, 于是 $r(A^*)=0$ 或 $r(A^*)=1$, 因为 $A^* \neq O$, 所以 $r(A^*)=1$, 于是 $r(A)=n-1$.

故齐次线性方程组 $AX=0$ 的基础解系含一个线性无关的解向量, 应选(B).

(14)【答案】 (C).

【解】 由 $P\{|X|<x\}=1-P\{|X|>x\}=1-2P\{X>x\}=\alpha$, 得 $P\{X \geq x\}=\frac{1-\alpha}{2}$.

再由 $P\{X > u_\alpha\}=\alpha$, 得 $x=u_{\frac{1-\alpha}{2}}$, 应选(C).

三、解答题

(15)【解】 方法一

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{\cos^2 x}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{x + \sin x \cos x}{x} \cdot \frac{x - \sin x \cos x}{x^3} \\ &= \lim_{x \rightarrow 0} \left(1 + \frac{\sin x}{x} \cos x \right) \cdot \frac{x - \sin x \cos x}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{2x - 2 \sin x \cos x}{x^3} = \lim_{x \rightarrow 0} \frac{2x - \sin 2x}{x^3} = 8 \lim_{x \rightarrow 0} \frac{2x - \sin 2x}{(2x)^3} \\ &\stackrel{2x=t}{=} 8 \lim_{t \rightarrow 0} \frac{t - \sin t}{t^3} = \frac{8}{3} \lim_{t \rightarrow 0} \frac{1 - \cos t}{t^2} = \frac{4}{3}. \end{aligned}$$

方法二

$$\begin{aligned} \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{\cos^2 x}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^2 \sin^2 x} = \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{x + \sin x \cos x}{x} \cdot \frac{x - \sin x \cos x}{x^3} \\ &= 2 \lim_{x \rightarrow 0} \frac{x - \sin x \cos x}{x^3} = 2 \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2} \\ &= \frac{2}{3} \lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2} = \frac{4}{3}. \end{aligned}$$

方法三

$$\begin{aligned}
 \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{\cos^2 x}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^2 \sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^4} = \lim_{x \rightarrow 0} \frac{x^2 - \frac{1}{4} \sin^2 2x}{x^4} \\
 &= \lim_{x \rightarrow 0} \frac{2x - \sin 2x \cos 2x}{4x^3} = \lim_{x \rightarrow 0} \frac{4x - \sin 4x}{8x^3} \\
 &= 8 \lim_{x \rightarrow 0} \frac{4x - \sin 4x}{(4x)^3} \stackrel{4x=t}{=} 8 \lim_{t \rightarrow 0} \frac{t - \sin t}{t^3} \\
 &= 8 \lim_{t \rightarrow 0} \frac{1 - \cos t}{3t^2} = \frac{4}{3}.
 \end{aligned}$$

方法四

$$\begin{aligned}
 \lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{\cos^2 x}{x^2} \right) &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^2 \sin^2 x} \\
 &= \lim_{x \rightarrow 0} \frac{x^2 - \sin^2 x \cos^2 x}{x^4} = \lim_{x \rightarrow 0} \frac{x^2 - \frac{1}{4} \sin^2 2x}{x^4} \\
 &= \lim_{x \rightarrow 0} \frac{x^2 - \frac{1 - \cos 4x}{8}}{x^4} = \frac{1}{8} \lim_{x \rightarrow 0} \frac{8x^2 - 1 + \cos 4x}{x^4},
 \end{aligned}$$

由 $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + o(x^4)$ 得 $\cos 4x = 1 - 8x^2 + \frac{32}{3}x^4 + o(x^4)$,

于是 $8x^2 - 1 + \cos 4x \sim \frac{32}{3}x^4$, 故 $\lim_{x \rightarrow 0} \left(\frac{1}{\sin^2 x} - \frac{\cos^2 x}{x^2} \right) = \frac{4}{3}$.

(16)【解】 如右图所示,

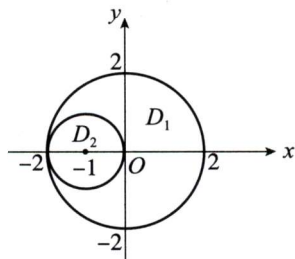
方法一 令 $D_1: x^2 + y^2 \leq 4$, $D_2: (x+1)^2 + y^2 \leq 1$, 则

$$\begin{aligned}
 \iint_D (\sqrt{x^2 + y^2} + y) dx dy &= \iint_{D_1} (\sqrt{x^2 + y^2} + y) dx dy - \\
 &\quad \iint_{D_2} (\sqrt{x^2 + y^2} + y) dx dy,
 \end{aligned}$$

$$\begin{aligned}
 \text{而 } \iint_{D_1} (\sqrt{x^2 + y^2} + y) dx dy &= \iint_{D_1} \sqrt{x^2 + y^2} dx dy \\
 &= \int_0^{2\pi} d\theta \int_0^2 r^2 dr = \frac{16\pi}{3},
 \end{aligned}$$

$$\begin{aligned}
 \iint_{D_2} (\sqrt{x^2 + y^2} + y) dx dy &= \iint_{D_2} \sqrt{x^2 + y^2} dx dy = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_0^{-2\cos\theta} r^2 dr \\
 &= -\frac{8}{3} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} \cos^3 \theta d\theta = \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^3 \theta d\theta = \frac{16}{3} \int_0^{\frac{\pi}{2}} \cos^3 \theta d\theta = \frac{32}{9},
 \end{aligned}$$

$$\text{于是 } \iint_D (\sqrt{x^2 + y^2} + y) dx dy = \frac{16\pi}{3} - \frac{32}{9}.$$



三(16)题图

方法二 令 $D_1 = \{(x, y) \mid x^2 + y^2 \leq 4, x \geq 0\}$,

$$D_2 = \{(x, y) \mid -2x \leq x^2 + y^2 \leq 4, x \leq 0\},$$

$$\text{而 } \iint_{D_1} (\sqrt{x^2 + y^2} + y) dx dy = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^2 (r + r \sin \theta) r dr = \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 + \sin \theta) d\theta = \frac{8}{3} \pi;$$

$$\iint_{D_2} (\sqrt{x^2 + y^2} + y) dx dy = \iint_{D_2} \sqrt{x^2 + y^2} dx dy = \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} d\theta \int_{-2\cos \theta}^2 r^2 dr$$

$$= \frac{8}{3} \int_{\frac{\pi}{2}}^{\frac{3\pi}{2}} (1 + \cos^3 \theta) d\theta$$

$$\stackrel{\theta - \pi = t}{=} \frac{8}{3} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (1 - \cos^3 t) dt$$

$$= \frac{16}{3} \int_0^{\frac{\pi}{2}} (1 - \cos^3 t) dt = \frac{16}{3} \left(\frac{\pi}{2} - \frac{2}{3} \right) = \frac{8}{3} \pi - \frac{32}{9},$$

$$\text{所以 } \iint_D (\sqrt{x^2 + y^2} + y) dx dy = \iint_{D_1} (\sqrt{x^2 + y^2} + y) dx dy + \iint_{D_2} (\sqrt{x^2 + y^2} + y) dx dy \\ = \frac{16}{3} \pi - \frac{32}{9}.$$

(17)【证明】 由 $\int_a^x f(t) dt \geq \int_a^x g(t) dt$, 得 $\int_a^x [f(t) - g(t)] dt \geq 0$,

$$\text{从而 } \int_a^b dx \int_a^x [f(t) - g(t)] dt \geq 0,$$

$$\text{而 } \int_a^b dx \int_a^x [f(t) - g(t)] dt = \int_a^b dt \int_t^b [f(t) - g(t)] dx = \int_a^b (b - t) [f(t) - g(t)] dt \\ = b \left[\int_a^b f(t) dt - \int_a^b g(t) dt \right] - \left[\int_a^b t f(t) dt - \int_a^b t g(t) dt \right] \\ = - \left[\int_a^b t f(t) dt - \int_a^b t g(t) dt \right] \geq 0,$$

$$\text{故 } \int_a^b t f(t) dt \leq \int_a^b t g(t) dt, \text{ 即 } \int_a^b x f(x) dx \leq \int_a^b x g(x) dx.$$

$$(18) \text{【解】} \quad (\text{I}) E_d = -\frac{P}{Q} \cdot \frac{dQ}{dP} = -(-5) \times \frac{P}{100 - 5P} = \frac{P}{20 - P}.$$

$$(\text{II}) R = PQ, \quad \frac{dR}{dP} = Q + P \frac{dQ}{dP} = Q - 5P,$$

$$\text{又 } Q(1 - E_d) = Q - 5P, \text{ 所以 } \frac{dR}{dP} = Q(1 - E_d),$$

若价格降低收益增加, 则 $\frac{dR}{dP} < 0$, 故 $1 - E_d < 0$, 即 $1 - \frac{P}{20 - P} < 0$, 解得 $P > 10$.

注意到 $0 < P < 20$, 故当 $10 < P < 20$ 时, 价格降低反而使收益增加.

$$(19) \text{【解】} \quad (\text{I}) \text{ 令 } S(x) = \frac{x^4}{2 \times 4} + \frac{x^6}{2 \times 4 \times 6} + \frac{x^8}{2 \times 4 \times 6 \times 8} + \cdots, \text{ 则}$$

$$S'(x) = \frac{x^3}{2} + \frac{x^5}{2 \times 4} + \frac{x^7}{2 \times 4 \times 6} + \cdots = \frac{x^3}{2} + xS(x),$$

$$\text{故 } S(x) \text{ 满足的微分方程为 } S'(x) - xS(x) = \frac{x^3}{2}.$$

(II) 由 $S'(x) - xS(x) = \frac{x^3}{2}$, 得

$$S(x) = \left(\int \frac{x^3}{2} e^{\int -x dx} dx + C \right) e^{-\int -x dx} = \left(\int \frac{x^3}{2} e^{-\frac{x^2}{2}} dx + C \right) e^{\frac{x^2}{2}} = C e^{\frac{x^2}{2}} - \frac{x^2}{2} - 1,$$

由 $S(0) = 0$, 得 $C = 1$, 故 $S(x) = e^{\frac{x^2}{2}} - \frac{x^2}{2} - 1$.

(20) 【解】 方法一 β 可否由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示等价于非齐次线性方程组 $x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta$ 是否有解.

$$\begin{aligned} \bar{A} = (\alpha_1, \alpha_2, \alpha_3, \beta) &= \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & a+2 & -b-2 & 3 \\ 0 & -3a & a+2b & -3 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & a & -b & 1 \\ 0 & -3a & a+2b & -3 \end{array} \right) \\ &\rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & a & -b & 1 \\ 0 & 0 & a-b & 0 \end{array} \right). \end{aligned}$$

(I) 当 $a = 0$, b 为任意常数时, 因为 $r(A) \neq r(\bar{A})$, 所以 β 不可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示;

(II) 当 $a \neq 0$ 且 $a \neq b$ 时, β 可由 $\alpha_1, \alpha_2, \alpha_3$ 唯一线性表示,

$$\text{由 } \bar{A} \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{1}{a} \\ 0 & 1 & 0 & \frac{1}{a} \\ 0 & 0 & 1 & 0 \end{array} \right), \text{ 得 } \beta = \left(1 - \frac{1}{a}\right)\alpha_1 + \frac{1}{a}\alpha_2 + 0\alpha_3.$$

(III) 当 $a \neq 0$ 且 $a = b$ 时,

$$\text{由 } \bar{A} \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{a} \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{1}{a} \\ 0 & 1 & -1 & \frac{1}{a} \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$\text{得 } x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta \text{ 的通解为 } X = C \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 - \frac{1}{a} \\ \frac{1}{a} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{a} \\ C + \frac{1}{a} \\ C \end{pmatrix} \quad (C \text{ 为任意常数}),$$

故 $\beta = \left(1 - \frac{1}{a}\right)\alpha_1 + \left(C + \frac{1}{a}\right)\alpha_2 + C\alpha_3$ (C 为任意常数).

方法二 令 $A = (\alpha_1, \alpha_2, \alpha_3)$, $\bar{A} = (\alpha_1, \alpha_2, \alpha_3, \beta)$,

$$|\alpha_1, \alpha_2, \alpha_3| = \begin{vmatrix} 1 & 1 & -1 \\ 2 & a+2 & -b-2 \\ 0 & -3a & a+2b \end{vmatrix} = \begin{vmatrix} 1 & 1 & -1 \\ 0 & a & -b \\ 0 & -3a & a+2b \end{vmatrix} = a(a-b).$$

(I) 当 $a = 0$ 时,

$$\bar{A} = \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 2 & 2 & -b-2 & 3 \\ 0 & 0 & 2b & -3 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & -b & 1 \\ 0 & 0 & 2b & -3 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 0 & -b & 1 \\ 0 & 0 & 0 & -1 \end{array} \right),$$

则当 $a=0$, b 为任意常数时, 因为 $r(\mathbf{A}) \neq r(\overline{\mathbf{A}})$, 所以 β 不可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示;

当 $a \neq 0$ 时, $\overline{\mathbf{A}} \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & a & -b & 1 \\ 0 & 0 & a-b & 0 \end{array} \right)$, 则

(II) 当 $a \neq 0$ 且 $a \neq b$ 时, 因为 $r(\mathbf{A}) = r(\overline{\mathbf{A}}) = 3$, 所以 β 可由 $\alpha_1, \alpha_2, \alpha_3$ 唯一线性表示,

$$\text{由 } \overline{\mathbf{A}} \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & a & -b & 1 \\ 0 & 0 & a-b & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & 0 & 1 \\ 0 & a & 0 & 1 \\ 0 & 0 & 1 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{1}{a} \\ 0 & 1 & 0 & \frac{1}{a} \\ 0 & 0 & 1 & 0 \end{array} \right),$$

$$\text{得 } \beta = \left(1 - \frac{1}{a}\right)\alpha_1 + \frac{1}{a}\alpha_2 + 0\alpha_3;$$

(III) 当 $a \neq 0$ 且 $a = b$ 时, $r(\mathbf{A}) = r(\overline{\mathbf{A}}) < 3$, β 可由 $\alpha_1, \alpha_2, \alpha_3$ 线性表示, 但表示方法不唯一.

$$\text{由 } \overline{\mathbf{A}} \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & a & -a & 1 \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 1 & -1 & 1 \\ 0 & 1 & -1 & \frac{1}{a} \\ 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 - \frac{1}{a} \\ 0 & 1 & -1 & \frac{1}{a} \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$\text{得方程组 } x_1\alpha_1 + x_2\alpha_2 + x_3\alpha_3 = \beta \text{ 的通解为 } \mathbf{X} = C \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 - \frac{1}{a} \\ \frac{1}{a} \\ 0 \end{pmatrix} = \begin{pmatrix} 1 - \frac{1}{a} \\ C + \frac{1}{a} \\ C \end{pmatrix} \quad (C \text{ 为任意常数}),$$

$$\text{故 } \beta = \left(1 - \frac{1}{a}\right)\alpha_1 + \left(C + \frac{1}{a}\right)\alpha_2 + C\alpha_3 \quad (C \text{ 为任意常数}).$$

$$\begin{aligned} (21) \text{【解】} \quad (I) \text{ 由 } |\lambda \mathbf{E} - \mathbf{A}| &= \begin{vmatrix} \lambda - 1 & -b & \cdots & -b \\ -b & \lambda - 1 & \cdots & -b \\ \vdots & \vdots & \ddots & \vdots \\ -b & -b & \cdots & \lambda - 1 \end{vmatrix} \\ &= [\lambda - 1 - (n-1)b](\lambda - 1 + b)^{n-1} = 0, \end{aligned}$$

得 $\mathbf{A}$ 的特征值为 $\lambda_1 = 1 + (n-1)b$, $\lambda_2 = \lambda_3 = \cdots = \lambda_n = 1 - b$.

情形一: $b \neq 0$

当 $\lambda_1 = 1 + (n-1)b$ 时, 因为 $\mathbf{A}$ 为实对称矩阵, 所以 $\lambda_1 = 1 + (n-1)b$ 只有一个线性无关

$$\text{的特征向量, 注意到 } [1 + (n-1)b]\mathbf{E} - \mathbf{A} = \begin{pmatrix} (n-1)b & -b & \cdots & -b \\ -b & (n-1)b & \cdots & -b \\ \vdots & \vdots & \ddots & \vdots \\ -b & -b & \cdots & (n-1)b \end{pmatrix} \text{ 的每}$$

行元素不为零, 于是 $\lambda_1 = 1 + (n-1)b$ 对应的特征向量为 $\xi_1 = (1, 1, \cdots, 1)^T$.

当 $\lambda_2 = \lambda_3 = \cdots = \lambda_n = 1 - b$ 时,

$$(1-b)E-A = \begin{bmatrix} -b & -b & \cdots & -b \\ -b & -b & \cdots & -b \\ \vdots & \vdots & & \vdots \\ -b & -b & \cdots & -b \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix},$$

则 $\lambda = 1-b$ 对应的线性无关的特征向量为

$$\xi_2 = (-1, 1, 0, \cdots, 0)^T, \xi_3 = (-1, 0, 1, 0, \cdots, 0)^T, \cdots, \xi_n = (-1, 0, \cdots, 0, 1)^T;$$

情形二: $b=0$

此时 $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 1$, 任意非零向量皆为矩阵 A 的特征向量.

(II) 当 $b \neq 0$ 时, 取 $P = (\xi_1, \xi_2, \cdots, \xi_n)$, 则

$$P^{-1}AP = \begin{bmatrix} (n-1)b+1 & 0 & \cdots & 0 \\ 0 & 1-b & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1-b \end{bmatrix};$$

当 $b=0$ 时, 对于任意可逆矩阵 P , 有 $P^{-1}AP = E$.

(22)【解】 由 $P(B|A) = \frac{P(AB)}{P(A)}$, 得 $P(AB) = P(A)P(B|A) = \frac{1}{12}$.

又由 $P(A|B) = \frac{P(AB)}{P(B)}$, 得 $P(B) = \frac{P(AB)}{P(A|B)} = \frac{1}{6}$.

(I) (X, Y) 的可能取值为 $(0, 0)$, $(0, 1)$, $(1, 0)$, $(1, 1)$,

$$\begin{aligned} P\{X=0, Y=0\} &= P(\overline{A}\overline{B}) = P(\overline{A+B}) = 1 - P(A+B) \\ &= 1 - P(A) - P(B) + P(AB) = \frac{2}{3}; \end{aligned}$$

$$P\{X=0, Y=1\} = P(\overline{A}B) = P(B) - P(AB) = \frac{1}{12};$$

$$P\{X=1, Y=0\} = P(A\overline{B}) = P(A) - P(AB) = \frac{1}{6};$$

$$P\{X=1, Y=1\} = P(AB) = \frac{1}{12}.$$

Y	X	
	0	1
0	$\frac{2}{3}$	$\frac{1}{6}$
1	$\frac{1}{12}$	$\frac{1}{12}$

$$(II) X \sim \begin{pmatrix} 0 & 1 \\ \frac{3}{4} & \frac{1}{4} \end{pmatrix}, Y \sim \begin{pmatrix} 0 & 1 \\ \frac{5}{6} & \frac{1}{6} \end{pmatrix}, XY \sim \begin{pmatrix} 0 & 1 \\ \frac{11}{12} & \frac{1}{12} \end{pmatrix},$$

$$E(X) = \frac{1}{4}, E(X^2) = \frac{1}{4}, D(X) = E(X^2) - [E(X)]^2 = \frac{3}{16},$$

$$E(Y) = \frac{1}{6}, E(Y^2) = \frac{1}{6}, D(Y) = E(Y^2) - [E(Y)]^2 = \frac{5}{36},$$

由 $E(XY) = \frac{1}{12}$ 得 $\text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{1}{12} - \frac{1}{24} = \frac{1}{24}$,

$$\text{则 } \rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{D(X)} \cdot \sqrt{D(Y)}} = \frac{\frac{1}{24}}{\frac{\sqrt{3}}{4} \times \frac{\sqrt{5}}{6}} = \frac{\sqrt{15}}{15}.$$

(Ⅲ) $Z = X^2 + Y^2$ 的可能取值为 0, 1, 2,

$$P\{Z=0\} = P\{X=0, Y=0\} = \frac{2}{3},$$

$$P\{Z=1\} = P\{X=0, Y=1\} + P\{X=1, Y=0\} = \frac{1}{4},$$

$$P\{Z=2\} = P\{X=1, Y=1\} = \frac{1}{12},$$

$$\text{故 } Z \text{ 的分布律为 } Z \sim \begin{pmatrix} 0 & 1 & 2 \\ \frac{2}{3} & \frac{1}{4} & \frac{1}{12} \end{pmatrix}.$$

(23)【解】 (I) 当 $\alpha = 1$ 时, 随机变量 X 的密度函数为

$$f(x; \beta) = \begin{cases} \beta x^{-\beta-1}, & x > 1, \\ 0, & x \leq 1, \end{cases}$$

$$E(X) = \int_1^{+\infty} x f(x, \beta) dx = \beta \int_1^{+\infty} x^{-\beta} dx = \frac{\beta}{\beta-1},$$

令 $E(X) = \bar{X}$, 则 β 的矩估计量为 $\hat{\beta} = \frac{\bar{X}}{\bar{X} - 1}$.

(II) 当 $\alpha = 1$ 时, 似然函数为

$$L(x_1, x_2, \dots, x_n; \beta) = f(x_1; \beta) f(x_2; \beta) \cdots f(x_n; \beta) = \beta^n (x_1 x_2 \cdots x_n)^{-\beta-1},$$

其中 $x_i > 1 (i = 1, 2, \dots, n)$,

$$\text{取对数得 } \ln L = n \ln \beta - (\beta + 1) \sum_{i=1}^n \ln x_i,$$

$$\text{由 } \frac{d}{d\beta} \ln L = \frac{n}{\beta} - \sum_{i=1}^n \ln x_i = 0, \text{ 得 } \beta \text{ 的最大似然估计值为 } \hat{\beta} = \frac{n}{\sum_{i=1}^n \ln x_i},$$

$$\beta \text{ 的最大似然估计量为 } \hat{\beta} = \frac{n}{\sum_{i=1}^n \ln X_i}.$$

(III) 当 $\beta = 2$ 时, X 的密度函数为

$$f(x; \alpha) = \begin{cases} 2\alpha^2 x^{-3}, & x > \alpha, \\ 0, & x \leq \alpha, \end{cases}$$

似然函数为

$$L(x_1, x_2, \dots, x_n; \alpha) = f(x_1; \alpha) f(x_2; \alpha) \cdots f(x_n; \alpha) = 2^n \alpha^{2n} (x_1 x_2 \cdots x_n)^{-3},$$

其中 $x_i > \alpha (i = 1, 2, \dots, n)$.

$$\text{取对数得 } \ln L = n \ln 2 + 2n \ln \alpha - 3 \sum_{i=1}^n \ln x_i,$$

因为 $\frac{d}{d\alpha} \ln L = \frac{2n}{\alpha} > 0$, 即 L 为 α 的增函数, 所以 α 的最大似然估计值为 $\hat{\alpha} = \min_{1 \leq i \leq n} \{x_i\}$,
最大似然估计量为 $\hat{\alpha} = \min_{1 \leq i \leq n} \{X_i\}$.

方法点评: 参数的矩估计法即通过总体的原点矩与样本相应阶的原点矩相等求出参数的估计量, 如 $E(X) = \bar{X}$ 等.

参数的极大似然估计法分为两种类型:

(1) 总体为离散型

先求 $L(\theta) = P\{X_1 = x_1\}P\{X_2 = x_2\} \cdots P\{X_n = x_n\}$, 再由 $\frac{d}{d\theta} \ln L(\theta) = 0$ 求出参数 θ 的极大似然估计量.

(2) 总体为连续型

先求 $L(\theta) = f(x_1; \theta)f(x_2; \theta) \cdots f(x_n; \theta)$, 再由 $\frac{d}{d\theta} \ln L(\theta) = 0$ 求出参数 θ 的极大似然估计量.